
Aswan S K

Data Science & AI/ML

Trivandrum, Kerala, India 695306

Email: aswansk.contact@gmail.com | Phone: +91 8111836945

LinkedIn: [linkedin.com/in/aswan-sk-066654301](https://www.linkedin.com/in/aswan-sk-066654301)

Professional Summary

Biomedical Engineer turned AI/ML & Data Science enthusiast with hands-on experience in data analytics, machine learning, and healthcare innovations. Passionate about leveraging data insights and AI models to optimize healthcare decision-making. Proficient in Python, SQL, Tableau, Power BI, and core machine learning principles. Ready to apply my interdisciplinary background to create impactful AI-driven solutions in the healthcare sector.

Core Competencies

- Machine Learning & AI (Supervised & Unsupervised Learning, Deep Learning)
 - Data Analysis & Pre processing (Pandas, NumPy, SciPy)
 - Data Visualization (Tableau, Power BI, Matplotlib, Seaborn)
 - Natural Language Processing (NLP)
 - Data Engineering (Data Pipeline Development, SQL, NoSQL)
 - Big Data Technologies (Spark, Hive, Pig)
 - Model Evaluation & Tuning (Hyperparameter Tuning, Cross-Validation)
 - Database Management (MySQL, PostgreSQL, MS SQL Server)
 - Agile/Scrum methodology
 - Communication & Data Storytelling
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Technical Skills

- **Programming Languages:** Python (Pandas, NumPy, TensorFlow, PYTORCH, KERAS, SCIKIT-learn), R (ggplot2, caret), SQL (MySQL, PostgreSQL), Java (Hadoop), JavaScript, SAS
 - **Machine Learning:** k-means, Deep Learning (Neural Networks, CNNs, RNNs, GANs)
 - **Data Visualization:** Matplotlib, Seaborn, Tableau, Power BI
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Education

- **Bachelor of Engineering in Biomedical Engineering**
TKM Institute of Technology, Kerala University of Technology (KTU) – 2023
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Certifications & Training

- **Data Science & Machine Learning** – CB Tech Institute, Kochi (2023 - 2024)
 - **Introduction to TensorFlow for AI** – Coursera
 - **Power BI & Excel for Data Analysis** – Microsoft 365 Certification
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Relevant Projects

1. **Android Application for Sign Language Conversion & Communication**
 - Developed a Python-based app using Media Pip to convert text to Indian Sign Language.
 - Integrated SQL for data storage and unit testing (N Unit) for quality assurance.
 - Applied NLP for text recognition, enabling real-time communication for the hearing-impaired.
2. **Data Analysis & Marketing Strategy**
 - Analysed data from 25K active users to refine marketing strategies, reducing drop-off rates by 30% and doubling app engagement.
 - Improved ETL efficiency using Pentaho Kettle, reducing analysis time by one week.
3. **Retail Data Strategy**
 - Designed merchandising strategies, increasing grocery retailer sales by 25%.
 - Developed targeted marketing strategies for fashion retailers, generating \$1M in revenue.
4. **Tableau Dashboard Development**
 - Created a Tableau dashboard to visualize KPIs such as Monthly Recurring Revenue, saving 10 hours per week in manual reporting.
5. **Data Warehousing & Integrity**
 - Developed a data warehouse tool, reducing operational costs by 80% and improving data quality by 20%.
 - Reduced redundancies by 50% in data management processes.

6. Business Operations Optimization

- Developed a program to eliminate duplicate entries with 92% accuracy, improving resource allocation by 80%.
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Internships & Hands-On Experience

1. Biomedical Intern

NS Institute of Medical Science, Kollam, Kerala – 2022

- Analysed healthcare data to improve decision-making and enhance patient care.

2. AI & Data Science Training

CB Tech Institute, Kochi – 2024

- Processed 20TB of data using SQL, Tableau, and Excel, delivering real-time insights that boosted revenue by 63%.
- Conducted cluster analysis on 2K customer patterns, optimizing marketing campaigns by 72%.
- Streamlined data collection processes, exceeding Q2 KPIs by 82%.

Generative AI in Healthcare

- Applied AI models for predictive healthcare models and data augmentation.
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Soft Skills

1. Problem Solving

- Ability to break down complex problems into smaller, manageable parts.
- Creative thinking to explore different approaches and techniques for data analysis and model development.
- Critical thinking to identify patterns and relationships in data.
- Willingness to experiment with new tools, methodologies, and frameworks.
- Balancing time spent on data cleaning, model development, and validation.
- Innovating with new machine learning techniques or algorithm modifications.

2. Collaboration & Teamwork

- Working effectively with cross-functional teams, including engineers, business stakeholders, and product managers.
- Open to feedback and iterating on models and analysis based on team input.
- Collaborative mindset to learn from others and share knowledge.

3. Adaptability & Learning Agility

- Staying up-to-date with the fast-evolving fields of AI/ML and Data Science.
 - Flexibility to switch between tasks and handle dynamic project requirements.
 - Willingness to experiment with new tools, methodologies, and frameworks.
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Languages

- **English:** High Proficiency
 - **Malayalam:** Native Proficiency
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Interests

- AI in Healthcare Innovation
 - Machine Learning & Generative Models
 - Technological Advancements in Data Science
 - Sports, Traveling, and Script Analysis
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References

Available upon request.
